# Imitation Learning

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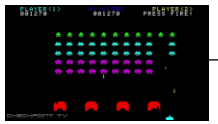
# Imitation Learning



RL

$a_1$ ?

$\boldsymbol{a_2}$ $\mathbf{0.3}$

$a_3$ ?

IL

$a_1$ 0

$\boldsymbol{a_2}$ $\mathbf{1}$ (expert choice)

$a_3$ 0

action chosen by expert

generated by expert

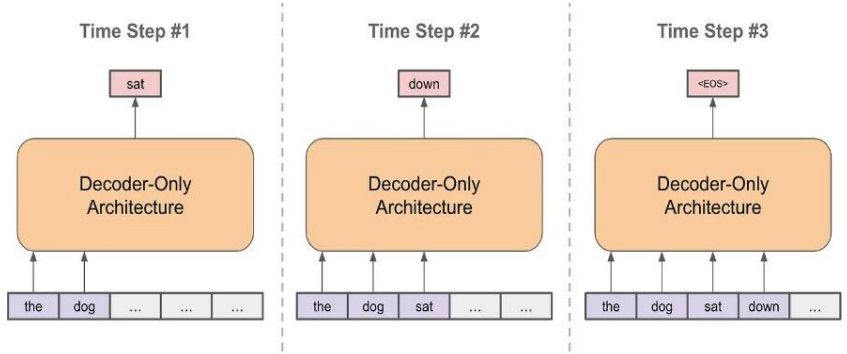
Offline IL: learn from static data generated by the expert $\{(s_1, a_1^\star, s_2, a_2^\star, \dots, s_H, a_H^\star)\}$

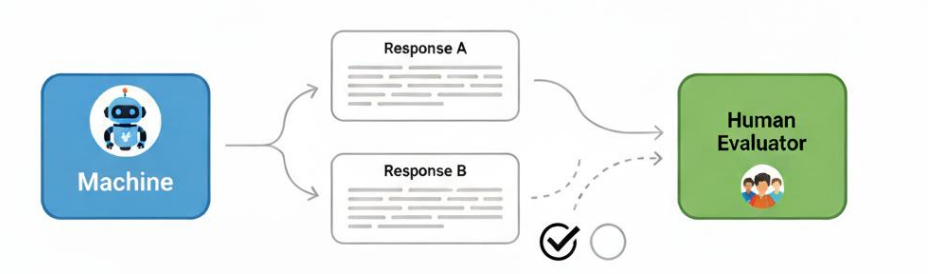
Online IL: may interact with MDP and query the expert $\{(s_1, a_1, a_1^\star, s_2, a_2, a_2^\star, \dots, s_H, a_H, a_H^\star)\}$

generated by the learner

# Examples

- Language models




- Robotics

# Types

- Direct Imitation:  directly learn policy to imitate the expert
  - Behavior cloning
  - DAgger
  - Direct preference optimization (preference feedback)
- Inverse RL:  learn reward function from expert, and perform RL on it
  - Generative Adversarial Imitation Learning (GAIL)
  - Adversarial Inverse Reinforcement Learning (AIRL)
  - RLHF (preference feedback)

# Direct Imitation

# Behavior Cloning: Reduction to Supervised Learning

**Behavior Cloning** (Offline IL)

Run expert policy $\pi^\star$ and obtain $(s_1, a_1), \ldots, (s_N, a_N)$

Train a policy $\pi_\theta$ such that

$$\pi_\theta(s_i) \approx a_i$$

(classification if actions are discrete) $\mathbb{1}\{\pi_\theta(s_i) \neq a_i\}$

(regression if actions are continuous) $\|\pi_\theta(s_i) - a_i\|^2$

# Behavior Cloning: Reduction to Supervised Learning

**Behavior Cloning** (Offline IL)

input output

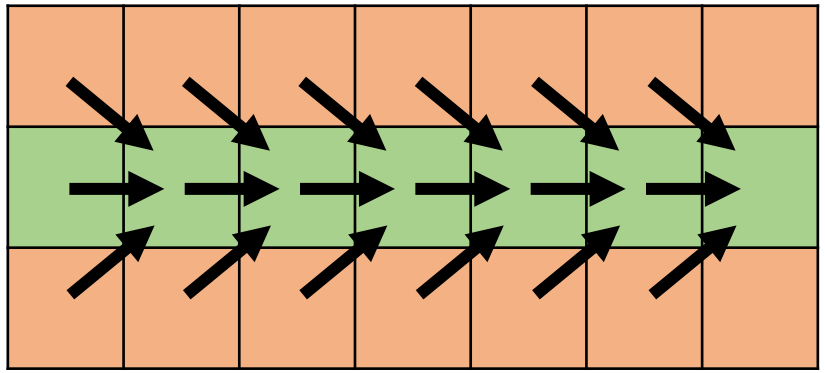
Run expert policy $\pi^\star$ and obtain $(s_1, a_1), \ldots, (s_N, a_N)$

Obtain policy $\pi_\theta$ by finding

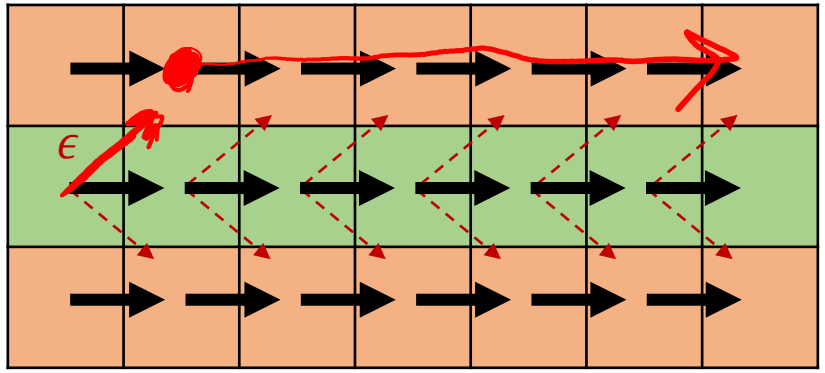
$$\theta = \underset{\theta}{\operatorname{argmin}} \ \frac{1}{N} \sum_{i=1}^{N} \log \frac{1}{\pi_\theta(a_i|s_i)} \qquad \text{(discrete action)}$$

$$\theta = \underset{\theta}{\operatorname{argmin}} \ \frac{1}{N} \sum_{i=1}^{N} \|a_i - \pi_\theta(s_i)\|^2 \qquad \text{(continuous action)}$$

# Behavior Cloning: Reduction to Supervised Learning



Optimal policy

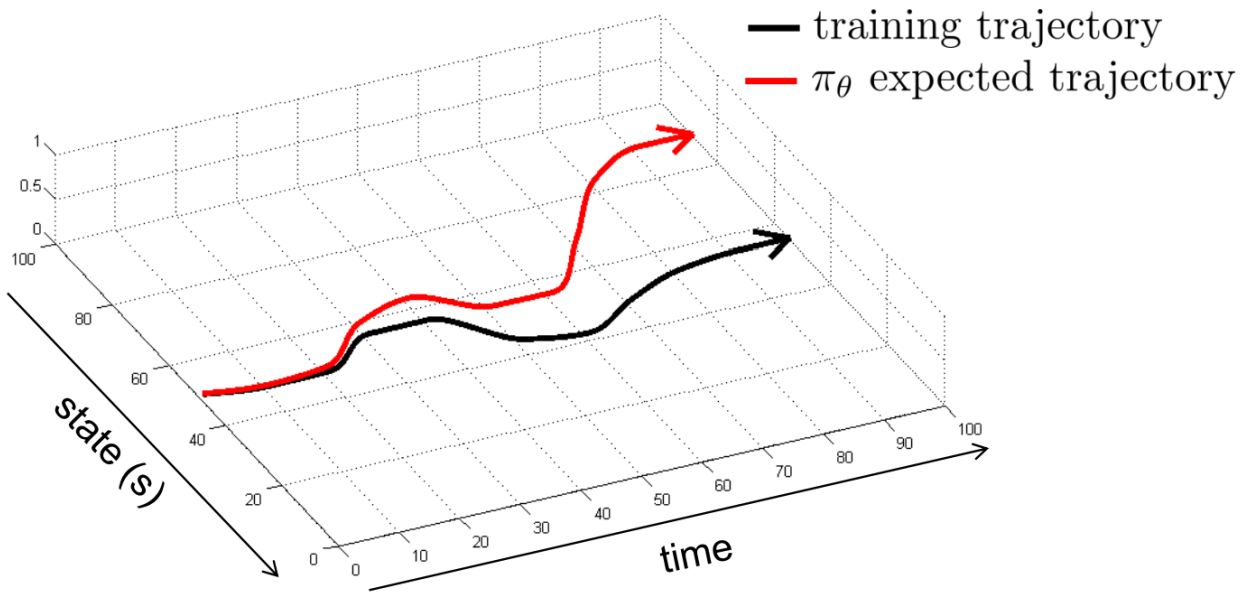



A policy with low training error ($\epsilon$)

but high testing error ($\approx \epsilon \times$ Horizon_length)

**Key issue:** the training data does not cover the states visited by the learner's policy

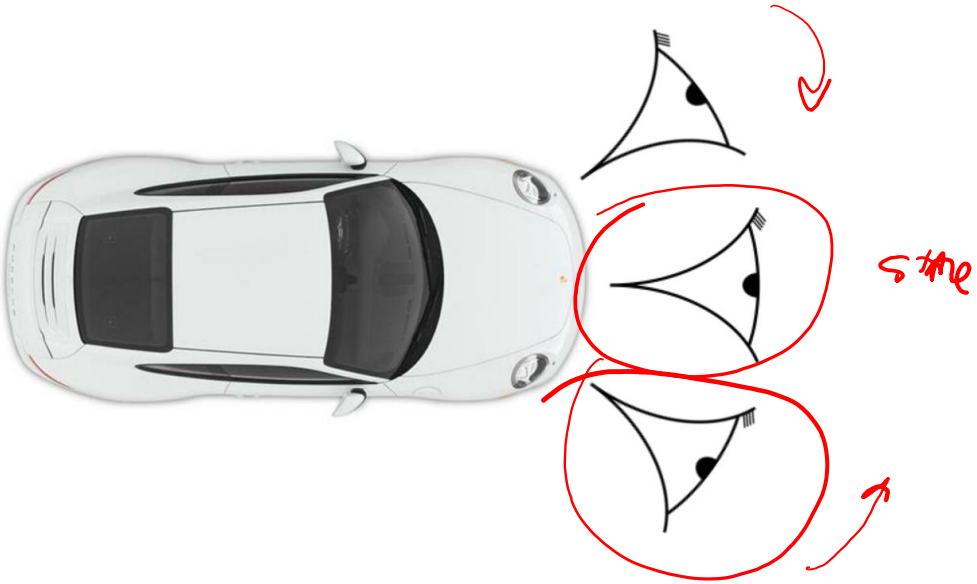
# Behavior Cloning: Reduction to Supervised Learning




Distribution shift

# Solution 1

- Data augmentation



Bojarsky et al. End to End Learning for Self-Driving Cars. 2016

https://www.youtube.com/watch?v=NJU9ULQUwng

# Solution 2: Interact with Expert (Online IL)

## Dataset Aggregation (DAgger)

For $k = 1,\ 2, \ldots$

Train $\pi_\theta(a|s)$ with dataset $\mathcal{B}$

Run $\pi_\theta$ to collect states $s_1, s_2, \ldots, s_N$

Ask the expert to label actions, giving $(s_1, a_1), \ldots, (s_N, a_N)$

Add $(s_1, a_1), \ldots, (s_N, a_N)$ to $\mathcal{B}$

Stephane Ross, Geoffrey J. Gordon, J. Andrew Bagnell. A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning

# Solution 2: Interact with Expert (Online IL)

## A Variant of DAgger

For $k = 1,\ 2, \ldots$

Train $\pi_\theta(a|s)$ with dataset $\mathcal{B}$

Run $\pi_\theta$

Ask the expert to take over at some time step

Store in $\mathcal{B}$ the samples generated by expert intervention

expert

$s_1, a_1, \ldots, s_m, a_m, s_{m+1}, \ldots$

Stephane Ross, Geoffrey J. Gordon, J. Andrew Bagnell. [A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning]

Key: collecting $(s, a)$ samples with $s$ generated from the learner (+ expert), and $a$ from the expert

input output

# Direct Imitation + RL

# Imitate then Reinforce

- A common paradigm of learning decision making:
  - Learn a policy $\pi_{\mathrm{BC}}$ through behavior cloning from experts
  - Run RL algorithm with the policy network initialized as $\pi_\theta = \pi_{\mathrm{BC}}$

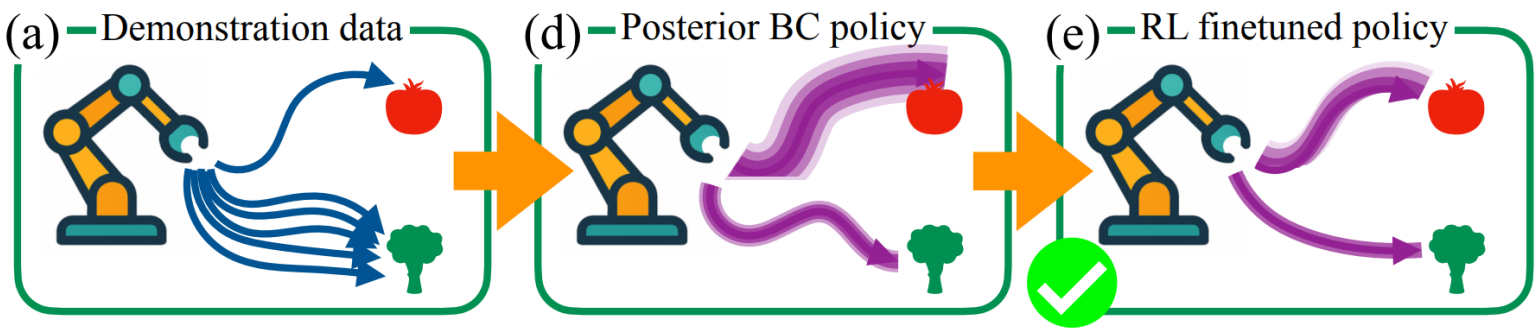
- This could largely save the time for RL to find the optimal policy
  - The amount / cost of exploration can be largely reduced

- Questions
  - What exploration strategy should the RL use?
  - How should we perform behavior cloning to facilitate downstream RL algorithm?

# Imitate and Reinforce

- For states where expert demonstration provides **sufficient** guidance
  - The learner does not need to explore that much
- For states where expert demonstration provides **insufficient** guidance
  - The learner may need to explore more

It's tempting to employ uncertainty-aware exploration in the RL phase, where the uncertainty accounts for the **uncertainty of the expert's action**.




Wagenmaker et al. Posterior Behavioral Cloning: Pretraining BC Policies for Efficient RL Finetuning. 2025.

# Imitate and Reinforce

$\pi_\theta(\cdot|s) = \mathcal{N}(\mu_\theta(s), \mathrm{cov})$

Run expert policy $\pi^\star$ and obtain $\mathcal{B} = \{(s_1, a_1), \dots, (s_N, a_N)\}$

Learn a policy $\mu_\theta$ through behavior cloning from $\mathcal{B}$

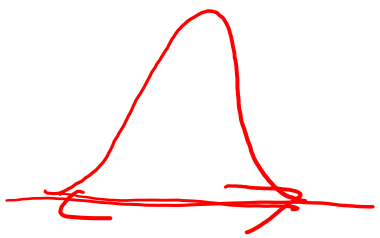
$\mu_\theta(s) \in \mathbb{R}^d$ (representing an action)

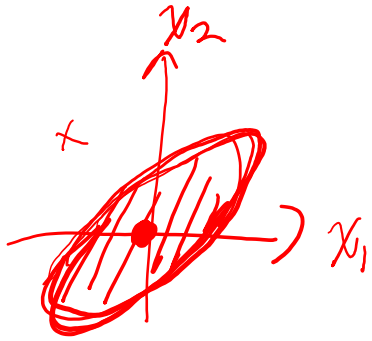
Run RL algorithm with policy initialized as $\mu_\theta$. When interacting with the environment, use policy

action $= (F_x, F_y, F_z, T_x, T_y, T_z)$

$$\mu_\theta(s) + \alpha \cdot \mathcal{N}(0, \mathrm{cov}(s))$$

Related to the action uncertainty on state $s$



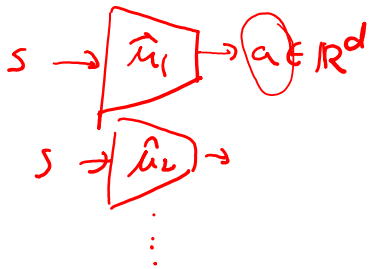



$\boxed{X \in \mathbb{R}^d}$

$\mathrm{cov} = \mathbb{E}\left[(X-\bar{X})(X-\bar{X})^T\right]$

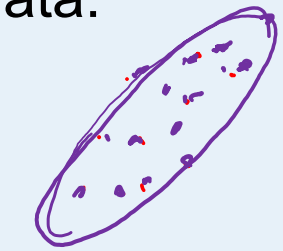
$\bar{X} = \mathbb{E}[X]$

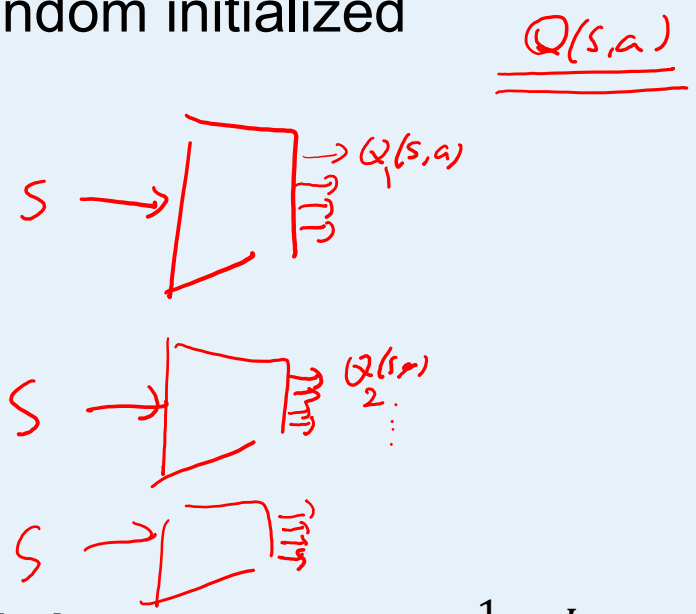
# How to Estimate Action Uncertainty




- Similar to techniques we discussed in the "Exploration" lecture, though there we mainly focused on the uncertainty of $Q^\star$ estimation
- The following idea is similar to Bootstrapped DQN (and also related to RND):

Train policies $\hat{\mu}_1, \hat{\mu}_2, \ldots, \hat{\mu}_L$ with Behavior Cloning with random initialized weights and randomly sub-sampled data:






For $\ell = 1, \ldots, L$:

Sample $\mathcal{B}_\ell \subset \mathcal{B}$ randomly

$$\hat{\mu}_\ell = \operatorname*{argmin}_{\mu} \sum_{(s,a)\in\mathcal{B}_\ell} \|\mu(s) - a\|^2$$

Define $\mathrm{cov}(s) = \frac{1}{L}\sum_{\ell=1}^{L} (\hat{\mu}_\ell(s) - \overline{\mu}(s))(\hat{\mu}_\ell(s) - \overline{\mu}(s))^\top \in \mathbb{R}^{d\times d}$ where $\overline{\mu}(s) = \frac{1}{L}\sum_{\ell=1}^{L} \hat{\mu}_\ell(s)$

# Imitate and Reinforce

Run expert policy $\pi^\star$ and obtain $\mathcal{B} = \{(s_1, a_1), \dots, (s_N, a_N)\}$

Learn a policy $\mu_\theta$ through behavior cloning from $\mathcal{B}$, and learn $\hat{\mu}_1, \hat{\mu}_2, \dots, \hat{\mu}_L$ using the procedure in the previous slide.

Run RL algorithm with policy initialized as $\mu_\theta$. When interacting with the environment, use policy

$$\mu_\theta(s) + \alpha \cdot \mathcal{N}(0, \mathrm{cov}(s))$$

Defined as in the previous slide

# Direct Imitation with Preference Feedback

# Preference Feedback from "Experts"

Offline

win lose

Given dataset $(x_1, a_{1w}, a_{1l}), (x_2, a_{2w}, a_{2l}), \dots (x_N, a_{Nw}, a_{Nl})$
where $a_{iw}$ is a **preferred action** by the expert than $a_{il}$ when seeing context $x_i$

Online

For $t = 1, \dots, T$:

Learner sees context $x_i$

Learner picks two actions $a_i^{(1)}, a_i^{(2)}$ and query the expert for their preference

With expert's feedback, the learner forms a tuple $(x_i, a_{iw}, a_{il})$

# Imitation with Preference Feedback

Direct Preference Optimization (DPO) (link) (An important technique for LLM alignment)

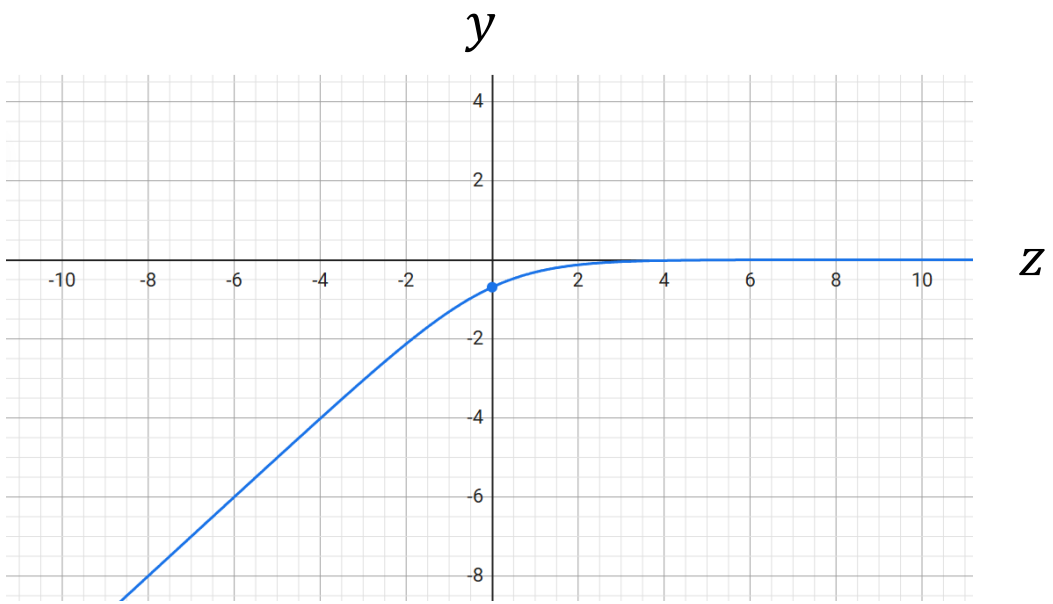
Given context $x$ and two potential actions, the expert chooses the **preferred** one (the preferred one is denoted as $y_w$ and the other is $y_l$).

$$\mathcal{L}_{\mathrm{DPO}}(\pi_\theta; \pi_{\mathrm{ref}}) = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y_w \mid x)}{\pi_{\mathrm{ref}}(y_w \mid x)} - \beta\log\frac{\pi_\theta(y_l \mid x)}{\pi_{\mathrm{ref}}(y_l \mid x)}\right)\right]$$

$\pi_{\mathrm{ref}}$ is a reference policy that $\pi_\theta$ is initialized as, and $\sigma(z) = \frac{1}{1+e^{-z}}$.

Ignoring $\pi_{\mathrm{ref}}$, DPO **maximizes** $\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y_w \mid x)}{\pi_\theta(y_l \mid x)}\right)\right]$

*c.f.* Behavior Cloning **maximizes** $\mathbb{E}_{(x,y^\star)\sim\mathcal{D}}[\log\pi_\theta(y^\star \mid x)]$




The function $y = \log \sigma(z)$

# Inverse Reinforcement Learning (Indirect Imitation)

# Inverse Reinforcement Learning

- Receive expert demonstration

  → **Infer the reward / value function** underlying the expert's behavior

  → Find a policy that optimizes over this reward function (RL)

- An indirect way of imitation learning

# Inverse RL

Given dataset $(s_1, a_1), (s_2, a_2), \ldots (s_N, a_N)$ where $a_i$'s are expert's actions,

can we **find a $\boldsymbol{Q}$-function** $Q(x, a)$ that explains the expert's behavior?

# Inverse RL

Find a reward / value function that explains the behavior of the expert.

- **Behavior assumption:** Assume a connections between expert policy and value:

$$\pi^\star(a|s) = \frac{\exp\big(Q^\star(s,a)\big)}{\sum_b \exp\big(Q^\star(s,b)\big)} \propto \exp\big(Q^\star(s,a)\big)$$

- **Maximum likelihood:** Given expert trajectories, find $\phi$ that maximizes likelihood

$$\phi = \underset{\phi}{\operatorname{argmax}} \sum_{i=1}^{N} \log \pi_\phi^\star(a_i|s_i) = \underset{\phi}{\operatorname{argmax}} \sum_{i=1}^{N} \log \frac{\exp(Q_\phi(s_i,a_i))}{\sum_b \exp\big(Q_\phi(s_i,b)\big)}$$

After finding this $\phi$, we may

If we do this, there is no difference with direct imitation

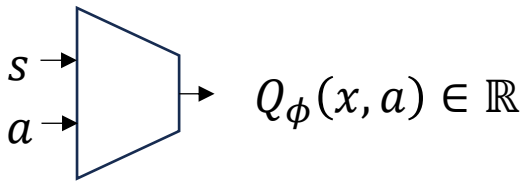
1) Directly get an induced policy $\pi(a|s) \propto \exp(Q_\phi(s,a))$, or

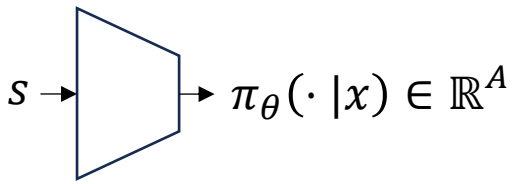
2) Train another policy $\pi_\theta(a|s)$ based on $Q_\phi(s,a)$

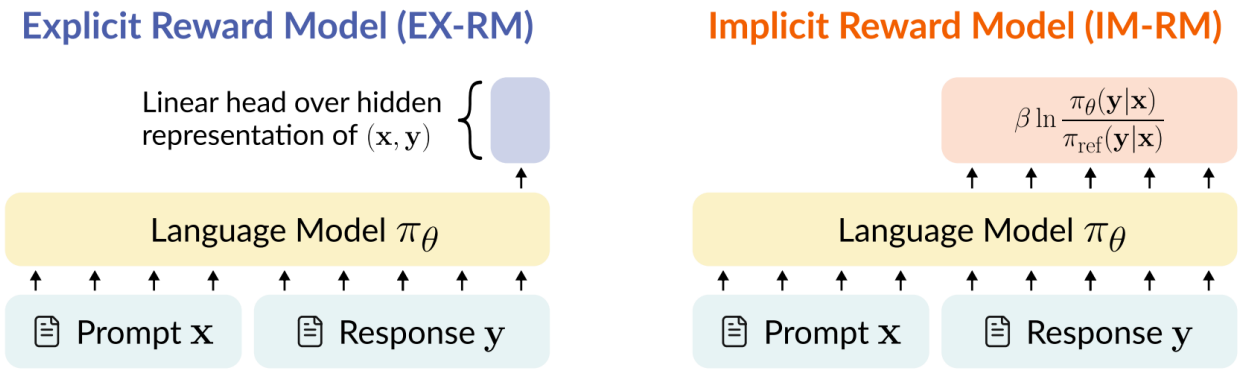
This is what makes IRL different from direct imitation

# Inverse RL

Inverse RL allows different parameterization of values and policies, e.g.




Razin et al., 2025. Why is Your Language Model a Poor Implicit Reward Model?

# Inverse RL with Preference Feedback

$$p^*(y_1 \succ y_2 \mid x) = \frac{\exp\left(r^*(x, y_1)\right)}{\exp\left(r^*(x, y_1)\right) + \exp\left(r^*(x, y_2)\right)}$$

Behavior assumption

$$\mathcal{L}_R(r_\phi, \mathcal{D}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}}\left[\log \sigma(r_\phi(x, y_w) - r_\phi(x, y_l))\right]$$

Maximum likelihood

$$\max_{\pi_\theta} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(y|x)}\left[r_\phi(x, y)\right] - \beta \mathbb{D}_{\mathrm{KL}}\left[\pi_\theta(y \mid x) \mid\mid \pi_{\mathrm{ref}}(y \mid x)\right]$$

Policy training

# Generative Adversarial Imitation Learning (GAIL)

- The previous approach performs two-phase training (training $Q_\phi$ → training $\pi_\theta$)
- It is also possible to **interleave** the training of $Q_\phi$ and $\pi_\theta$

# Generative Adversarial Imitation Learning (GAIL)

Recall the IRL paradigm:

- **Behavior assumption:** $\pi^\star(a|s) \propto \exp\big(Q^\star(s,a)\big)$
- **Maximum likelihood:** $\phi = \underset{\phi}{\operatorname{argmax}} \sum_{i=1}^{N} \log \pi_\phi^\star(a_i|s_i) = \underset{\phi}{\operatorname{argmax}} \sum_{i=1}^{N} \log \frac{\exp(Q_\phi(s_i,a_i))}{\sum_b \exp\big(Q_\phi(s_i,b)\big)}$
- **Induced policy / policy training:** $\pi(a|s) \propto \exp\Big(Q_\phi(s,a)\Big)$

Let's take a deeper look into the Maximum Likelihood step:

Entropy

$$\log \frac{\exp(Q_\phi(s_i,a_i))}{\sum_b \exp\big(Q_\phi(s_i,b)\big)} = Q_\phi(s_i,a_i) - \max_\pi \left( \sum_a \pi(a|s_i)\, Q_\phi(s_i,a) + H(\pi(\cdot\,|s_i)) \right)$$

$$\approx Q_\phi(s_i,a_i) - \max_\pi \left( \sum_a \pi(a|s)\, Q_\phi(s_i,a) \right)$$

(Let's ignore the entropy term…)

# Generative Adversarial Imitation Learning (GAIL)

This indicates an interleaved version of Inverse RL:

**Input:** expert data $(s_1, a_1), (s_2, a_2), \ldots (s_N, a_N)$

While True:

$$\phi \leftarrow \phi + \alpha \nabla_\phi \frac{1}{N} \sum_{i=1}^{N} \left( Q_\phi(s_i, a_i) - \sum_a \pi_\theta(a|s_i)\, Q_\phi(s_i, a) \right)$$

Value Training

$$\theta \leftarrow \theta + \alpha' \nabla_\theta \frac{1}{N} \sum_{i=1}^{N} \sum_a \pi_\theta(a|s_i)\, Q_\phi(s_i, a)$$

Policy Training

# Generative Adversarial Imitation Learning (GAIL)

$\phi$ tries to maximize $$\frac{1}{N}\sum_{i=1}^{N}\left(Q_\phi(s_i, a_i) - \sum_a \pi_\theta(a|s_i)\, Q_\phi(s_i, a)\right)$$

**Interpretation:** Find a value function $Q_\phi$ that makes

- the expert have high value
- the learner's policy $\pi_\theta$ have low value

$\theta$ tries to maximize $$\frac{1}{N}\sum_{i=1}^{N}\sum_a \pi_\theta(a|s_i)\, Q_\phi(s_i, a)$$

**Interpretation:** Train a policy so that $\pi_\theta$ gains high value

# Summary for Imitation Learning

- Direct Imitation
  - Behavior Cloning
  - Dagger
  - Behavior Cloning + RL with refined exploration bonus
- Indirect Imitation (Inverse RL)
  - Two-phase training
  - Interleaved training (GAIL, AIRL)